

Voltammetric Characterization of Grafted Polymer Modified with ZnO Nanoparticles on Glassy Carbon Electrode¹

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Received July 5, 2016; in final form, November 17, 2016

Abstract—In this study, a grafted polymer (GP) with ZnO nanoparticles (GP/ZnO NPs) was attached on the surface of glassy carbon electrode (GCE), in order to produce a new modified electrode (GP/ZnO NPs-GCE). The gamma irradiation method was used to grafted polystyrene (polymer) with acrylonitrile (monomer), while slow evaporation process was used to prepare the new modified electrode. The cyclic voltammetry (CV) of $K_4[Fe(CN)_6]$ was used to study the electrochemical properties GP/ZnO NPs-GCE. The peak separation (ΔE_{pa-c}) was 500 mV between the redox peaks of Fe(II)/Fe(III) in an aqueous solution of 1 M KCl and the current ratio of redox current peaks (I_{pa}/I_{pc}) was ≈ 1 for the modified electrode. This indicated that the modified electrode has a good reversibility and conductivity, wherefore; it was applied in the voltammetric filed. It was found that the modified electrode GP/ZnO NPs-GCE have a reasonable solubility and stability at various pH medium. Additionally, the sensitivity of the electrochemical analysis by cyclic voltammetric (CV) method is extensively subjected to the pH medium and the scan rate (SR). A couple of redox current peaks of $K_4[Fe(CN)_6]$ in KCl solution was observed with a reversible process: Fe^{3+}/Fe^{2+} . Finally a good diffusion coefficient of electroactive species (D) for the new modified electrode was found in this study by chronoamperometry method using Cottrell equation.

Keywords: cyclic voltammetry, GP/ZnO NPs, KCl, Fe(II)/Fe(III), $K_4[Fe(CN)_6]$

DOI: 10.1134/S1023193518010068

INTRODUCTION

Many studies in electrochemistry of cyclic voltammetric method showed that the conductive of graft polymer improved by modification on the solid bar electrodes [1–6].

The electrochemical behavior of many structures such as Cu^{2+} , Fe^{3+}/Fe^{2+} redox couple, indium tin oxide (ITO) and ZnO nanoparticles (NPs) have been studied at modified Pt electrode. It was shown clearly that the rate of diffusion coefficient of electroactive species (D) and heterogeneous charge transfer constant increased for the modified electrode than the bare. For that reason ZnO NPs applied as biosensors for electrophile and nucleophile compounds existed in water [7].

Synthesize of advanced electrode material such as ZnO-graphene nanocomposites was demand for high-execution supercapacitors application. A facile method was used to synthesis ZnO-graphene nanocomposites by in situ wet chemistry process. The process including the formation of ZnO-graphene nanocomposites as

sandwich-structured by the embedded of high dispersed ZnO NPs in graphene nanosheets [8].

The anodic performance of ZnO was achieved in the mixture solution of ethanol, LiCl and water. A simple method was used to polarize ZnO NPs by cyclic voltammetry and chronoamperometry techniques under a given conditions [9, 10].

ZnO nanoparticles were used with polystyrene and polyvinylidene fluoride as a film deposited on electrode to determine electrochemical properties [11, 12]. Other studies on the utilization of polymers with nanoparticles at the end of grafted polymer chain were characterized with different electrochemical properties [13].

A novel material with high-performance that used in numerous applications such as the industrial fields can be provided by the uses of inorganic nanoparticles embedded in the polymer matrix. New attractive applications such as optoelectronic and magnetooptic can be designed and fabricated by understanding the physics behind of novel functional polymer-inorganic nanocomposites [14].

¹ The article is published in the original.

Recently, an indium tin oxide electrode decorated with gold quantum dots covered with layer of polymer nanostructures was used as electrochemical biosensor [15]. The cyclic voltammetric analysis showed good electrochemical behavior of adenine and guanine in serum sample at low detection limit of 250 to 500 nM correspondingly.

In this study, electrode sensing elements of graft polystyrene-acrylonitrile modified with ZnO NPs were synthesized via slow evaporation technique on the surface of glassy carbon.

EXPERIMENTAL

Equipments and Electro-Analytical Analysis Methods

Equipments: Cyclic voltammetry (CV) was performing by using EZstat series NuVant systems Inc. pioneering electrochemical technologies USA. Platinum wire (1 mm diameter) was utilized to be a counter electrode while Ag/AgCl (3 M NaCl) was considered as a reference electrode. The modified glassy carbon effective electrode (GCE) was used in this study after cleaning with alumina grand. The surface morphologies of the samples were investigated by scanning electron microscope (SEM—JEOL) operated at 20–30 kV. In this work, the specimens were prepared for SEM study using a standard procedure as described in our previous researcher [16].

Procedure: cyclic voltammetric cell was used in this technique by adding 10 mL of electrolyte in the quartz cell and immerse three electrodes in the electrolyte medium (GP/ZnO NPs-GCE as working electrode, Ag/AgCl reference electrode and counter electrode), then these three electrodes was connected with potentiostat to finding the results by the cyclic voltammogram using personal computer.

Reagents

Zinc oxide (ZnO) nanoparticles purchased from Sigma Aldrich (>50 nm) (USA), Normal saline (0.9% NaCl w/v%) from Alcon Parenterals (India) Ltd. The deionize water was used to diluted all the solutions. An aqueous solution of 1 M of KCl was used as electrolyte supporting element. All materials were with purity of 99.9% thus they used without any further purification process.

Synthesis of Grafted Polymer (GP)

By means of gamma-irradiation technique a crylonitrile was used as a monomer to graft the polystyrene as a polymer. The solvent and catalyst used in this process were chloroform and ammonium iron(II) sulphate, respectively. Different percentages of the graft polymer were collected and studied [17].

Preparation of Grafted Polymer with ZnO Nanoparticles (GP/ZnO NPs)

The grafted polystyrene with acrylonitrile was modified with ZnO nanoparticles by dissolving 1 g of graft polymer (GP) in a good solvent such as chloroform and then mixing with 1 mg of ZnO NPs. The mixture was blended using magnetic stirrer to produce a homogenous mixture of graft polymer with ZnO NPs. Then, the solvent was evaporated by placing the mixture in open container at room temperature over night. The final obtained product was the remaining ZnO nanoparticles embedded into the grafted polystyrene (GP/ZnO NPs).

Preparation of GP/ZnO NPs Modified GCE

Fast evaporation method was used to cover the GCE with GP/ZnO NPs. In order to obtain an appropriate thickness of GP/ZnO NPs thin film on the GCE, two simple steps were followed. First 1 g of GP/ZnO NPs was dissolved in chloroform to obtain a homogeneous mixture. Then 1 μ L of the prepared solution was poured over the clean surface of GCE and then dried by an airbrush.

RESULTS AND DISCUSSION

Surface Morphology Study

The scanning electron microscopes (SEM) images are shown in Fig. 1. It is clear that the surface morphology of the as prepared polymer is homogeneous, dense, flat and uniform with no noticeable pores as shown in Fig. 1a. There is no verification was observed to indicate any damage or roughness for the surface of the polymeric material. Figure 1b shows the SEM image of the modified polymer with ZnO nanoparticle. The image revealed that the polymer was well modified with ZnO nanoparticles. As well as, it is obvious that the nanoparticles have homogeneous, compact spherical shape with 50 nm diameter and they are well embedded and distributed on the surface of grafted polymer.

The Effect of Various Supporting Electrolyte on the GP/ZnO NPs-GCE

The various supporting electrolytes showed as expected insignificant effect on the potential anodic peak of Fe(III), mainly in aqueous media. The values of anodic current peak (I_{pa}), anodic potential peak (E_{pa}), cathodic current peak (I_{pc}) and cathodic potential peak (E_{pc}) were illustrated in Table 1. Generally, among variety of supporting electrolytes Fe ions exhibited identical redox current peaks. Nevertheless, the utilization of KCl as electrolyte induced significant increment in oxidation current peak since GP/ZnO NPs-GCE was 3.7 times higher compared with GCE.

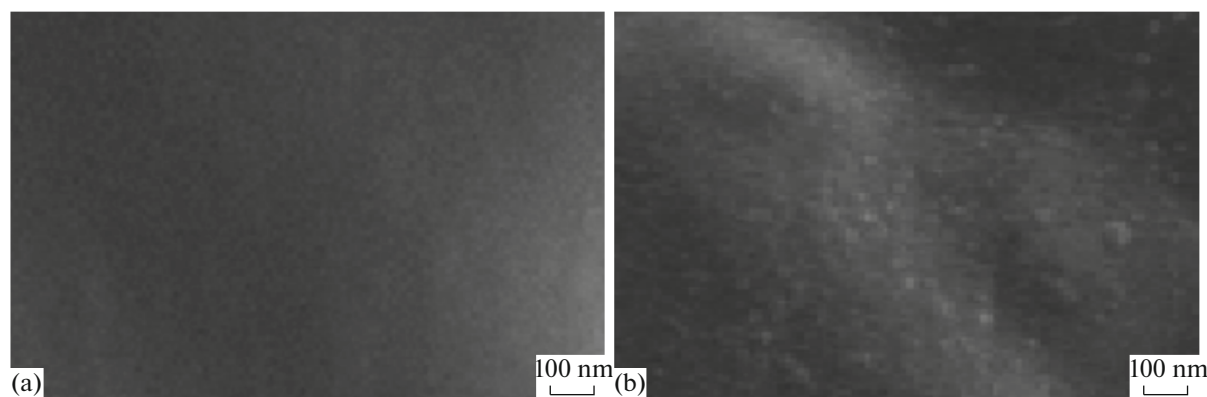


Fig. 1. The morphology grafted polymer (a) before and (b) after modification with ZnO nanoparticles.

Thus the utilization of KCl as an electrolyte in this study produced the highest current output.

Characterization of GP/ZnO NPs Using GCE

Calibration procedure for the cyclic voltammetric system in aqueous solutions was done by using $K_4[Fe(CN)_6]$ as a reference standard solution [18].

The cyclic voltammograms using a GP/ZnO NPs-GCE working electrode were shown in Fig. 2, while Fig. 3 show the cyclic voltammograms of GCE in KCl with $K_4[Fe(CN)_6]$.

The anodic current peak of the Fe^{2+}/Fe^{3+} appears to be extensively enhanced with 3.7 times, these according to the applied electrolyte with existence of modified GCE comparing with GCE, as illustrated in Table 1. The separation between the redox peaks (ΔE_{pa-c}) of Fe^{2+}/Fe^{3+} in an aqueous medium of KCl is

500 mV (for $n = 1$). While the ratio between the oxidation to reduction current peaks (I_{pa}/I_{pc}) is ≈ 1 for the modified working electrode with GP/ZnO NPs-GCE. These results represented the excellent reversibility and conductivity of the electrode comparing to previous studies [19].

Effect of the Scan Rate Versus GP/ ZnO NPs-GCE

The influence of varying scan rate (SR) on the peaks of redox current of $K_4[Fe(CN)_6]$ solution by cyclic voltammograms was investigated over a range of SR with $0.1-30 \text{ mV s}^{-1}$, where the used working electrode was GP/ ZnO NPs-GCE in electrolyte of 1 M KCl. It was observed that the redox current peaks of the $Fe(II)/Fe(III)$ couple increased with the increasing of SR, due to the IR effect and heterogeneous kinetics. The straight line was obtained from Fig. 4, which represent logarithm (I_{pa}) against logarithm

Table 1. Cyclic voltammetry of 1 mM $K_4[Fe(CN)_6]$ for the GCE and GP/ZnO NPs-GCE for a 100 mV/s scan rate at various electrolytes namely K_2HPO_4 , K_2SO_4 , Na_2SO_4 and normal saline

Electrolyte NPs/GCE	I_{pa} , μA GCE	E_{pa} , mV GCE	I_{pa} , μA GP-ZnO NPs/GCE	E_{pa} , mV GP-ZnO NPs/GCE	Enhancement
(Anodic)					
Normal saline	15	625	50	-1.23	3.3
K_2HPO_4	25	300	25	1.5	1
KCl	27	350	100	-1.25	3.7
K_2SO_4	30	300	2	-1.3	0.06
Na_2SO_4	30	300	70	-1.4	2.3
(Cathodic)					
Normal saline	-15	50	-110	-1.4	7.3
K_2HPO_4	-8	200	-5	500	0.63
KCl	-18	100	-200	-1.45	11
K_2SO_4	-12.5	200	-2.6	-1.55	0.21
Na_2SO_4	-12	200	-40	-1.5	3.3

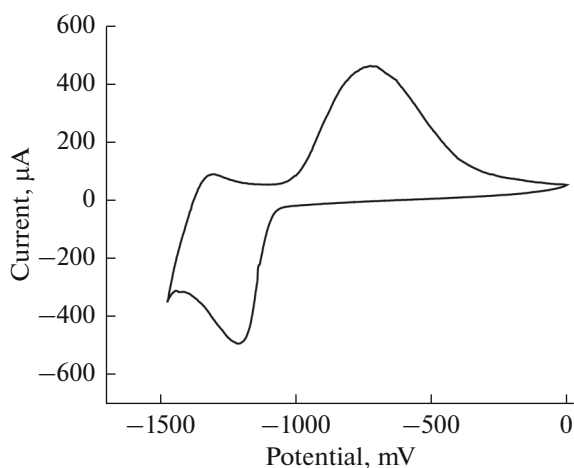


Fig. 2. Cyclic voltammogram of $K_4[Fe(CN)_6]$ in electrolyte KCl by working electrode GP/ZnO NPs-GCE against Ag/AgCl as reference electrode.

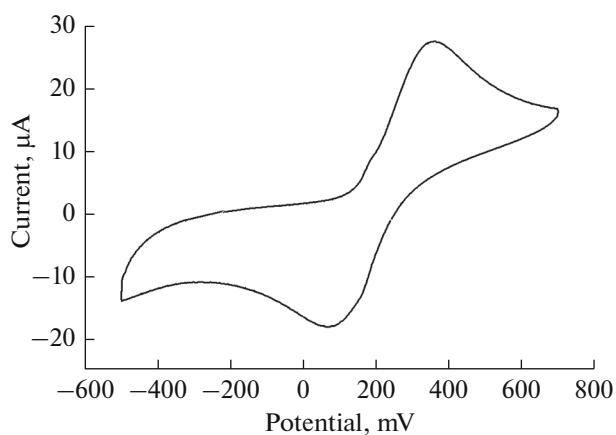


Fig. 3. Cyclic voltammogram of $K_4[Fe(CN)_6]$ in electrolyte KCl by working electrode GCE against Ag/AgCl as reference electrode.

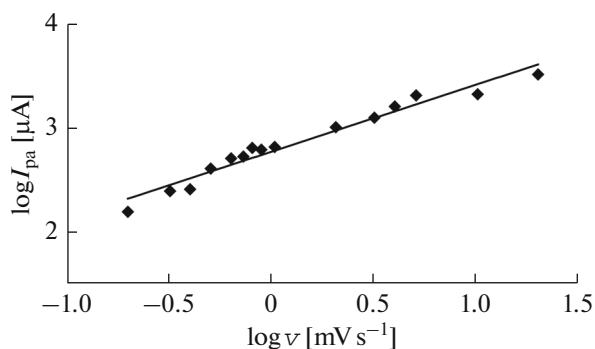


Fig. 4. $\log(I_{pa})$ of $K_4[Fe(CN)_6]$ in electrolyte KCl against $\log(SR)$ by working electrode GP/ZnO NPs-GCE at range of SR ($0.1\text{--}3\text{ mV s}^{-1}$).

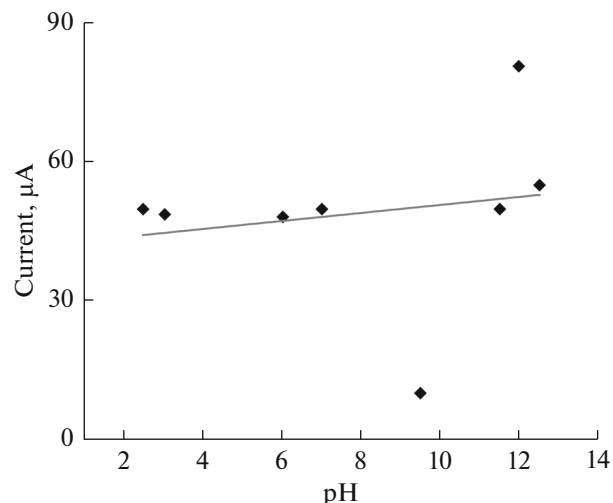


Fig. 5. Peak oxidation current variation with pH value (2–13) for $K_4[Fe(CN)_6]$ in electrolyte KCl by working electrode GP/ZnO NPs-GCE.

(SR) for the primary cycle of the oxidation current. It can be notice clearly that it fulfilling the mathematical relation $y = 0.6366x + 2.7685$ of good sensitivity $R^2 = 0.9589$.

The slope of 0.6 which obtained from the above equation show a close value with the theoretical slope of 0.5 for diffusion controlled process [20].

Voltammetric Characterization of GP/ZnO NPs-GCE at Different pH

The effect of pH value of the reaction solution was studied, where the pH value was controlled either by HCl or NaOH as shown in Fig. 5. It was noticed from the figure that the increasing of pH value (reduction acidity), raise the peaks of redox current of $K_4[Fe(CN)_6]$ at constant current value ($50\text{ }\mu\text{A}$). While

at the pH value of 9.5, decrement in current value of the oxidation process was observed. The alteration in the behavior of the plot, at pH (9.5), can be ascribed to the second dissociation constant of the $K_4[Fe(CN)_6]$. In addition, increasing the pH value to a higher value close to 12 raised the improvement of peaks of redox current.

Effect of the Concentration of $K_4[Fe(CN)_6]$ Using GP/ZnO NPs-GCE

Different concentration range of $K_4[Fe(CN)_6]$ with 2–10 mM was shown in Fig. 6. The result revealed that the linear current dependent on the concentration and the equation that describes this relationship is $y = 6.6786x + 85.714$ with $R^2 = 0.940$. Considerably high sensitivity response for $K_4[Fe(CN)_6]$

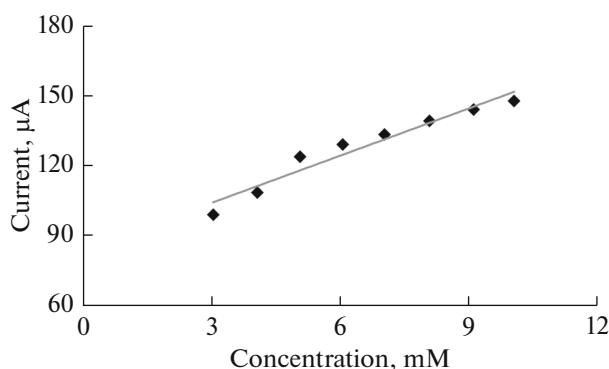


Fig. 6. The oxidation current of $K_4[Fe(CN)_6]$ in electrolyte KCl against $K_4[Fe(CN)_6]$ concentration by working electrode GP/ZnO NPs-GCE.

was observed from the slope of the linear line with $85.7 \mu A/mM$ for GP/ZnO NPs-GCE during the CV measurement.

Reliability and Stability of GP/ZnO NPs-GCE

The patterns of the redox current potential cycling was conducted meanwhile cyclic voltammetry for the modified working electrode GP/ZnO NPs-GCE in $K_4[Fe(CN)_6]$ with KCl at SR is 100 mV s^{-1} .

It was observed that the effect of redox current peak for the new modified working electrode differ than the continuous potential cycling, this is due to the effect of faradic activity is yet effective subsequent to several periods of potential cycling. Table 2 illustrated the reliability of current of the cathodic peak (I_{pc}) and the relative standard deviation (RSD) which it is $\pm 0.78\%$, while Table 3 illustrated the reliability for the current of the anodic peak (I_{pa}) and the RSD was calculated to be $\pm 0.8\%$.

Figure 7 illustrates the CV of peaks of redox current of $1 \text{ mM } K_4[Fe(CN)_6]$ in 1 M KCl at five times of $Fe(II)/Fe(III)$ which revealed a good stability of the CV of the modified GCE by overlapping of the voltammograms at five times.

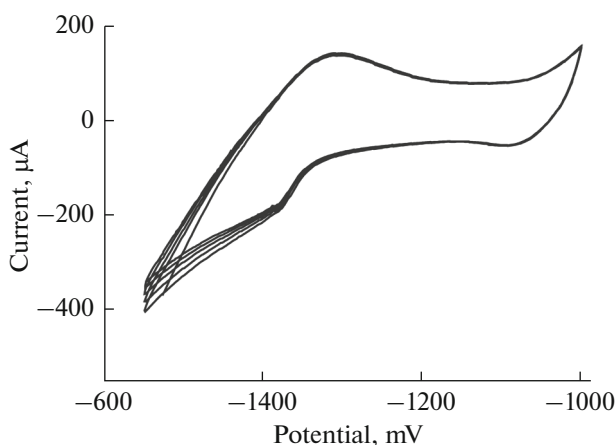


Fig. 7. The CV of five times for redox current peaks of GP/ZnO NPs-GCE in $K_4[Fe(CN)_6]$ with KCl at SR 100 mV s^{-1} .

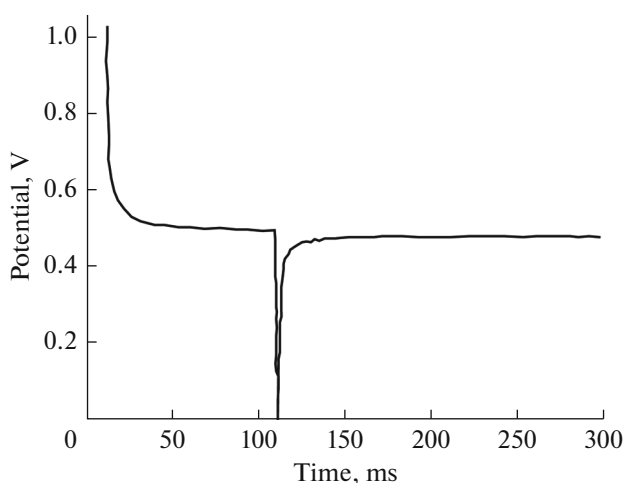


Fig. 8. Chronoamperogram or Cottrell draw for $K_4[Fe(CN)_6]$ in electrolyte KCl using GP/ZnO NPs-GCE against reference electrode Ag/AgCl.

Chronoamperometry Study

The repetitive increasing and decreasing of the current transient is in agreement with the theoretical prospect of Cottrell equation as shown in Fig. 8. The

Table 2. The reliability of GP/ZnO NPs-GCE as working electrode at SR 100 mV s^{-1} for cathodic peak $K_4[Fe(CN)_6]$ in electrolyte KCl

Numbers	$I_{pc}, \mu A$
1	200
2	201
3	202
4	203
5	204
Mean	202
RSD	$\pm 0.78\%$

Table 3. The reliability of GP/ZnO NPs-GCE as working electrode at SR is 100 mV s^{-1} for anodic peak $K_4[Fe(CN)_6]$ in electrolyte KCl

Numbers	$I_{pa}, \mu A$
1	199
2	198
3	197
4	196
5	195
Mean	197
RSD	$\pm 0.8\%$

potential was scanned in the inverse path from 0 to +1 V with pulse width of 300 ms. The D of the Fe(II)/Fe(III) was calculated depending on the diffusion progression to a reference electrode by using chronoamperometry method. It was found that D was equal to $9.3 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$, while it was equal to $5.1 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ when the GCE was used [19, 21, 22].

CONCLUSIONS

A graft polymer (GP) with ZnO nanoparticles was modified on the GCE surface to create a novel working modified electrode (GP/ZnO NPs-GCE). The new working electrode shows wide potential area at different pH conditions which can be used as a good sensitive nano-sensor with advantages such as easy preparation, non-toxic electrode and low cost. It was found that the sensitivity is affected highly by several parameters including concentration, pH value, type of electrolyte and scan rate.

The GP/ZnO NPs-GCE was show extremely stability when t five times voltammogram was used. The result shows the potential use of GP/ZnO NPs (GP/ZnO NPs-GCE) as a useful electrode material. Finally, good results of D were determined using chronoamperometry method by Cottrell equation for the new modified working electrode and the comparison with GCE.

REFERENCES

- Radhi, M.M., Tan, W.T., Ab Rahman, M.Z.B., and Kassim, A.B., Electrochemical characterization of the redox couple of $[\text{Fe}(\text{CN})_6]^{3-}/\text{Fe}(\text{CN})_6]^{4-}$ mediated by a graft polymer modified glassy carbon electrode, *J. Chem. Eng. Jpn.*, 2010, vol. 43, pp. 927–931.
- Labaye, D.E., Jerome, C., Gueskin, V.M., Louette, P., Lazzaroni, R., Martinot, L., and Jerome, R., Full electrochemical synthesis of conducting polymer films chemically graft to conducting surfaces, *J. Am. Chem. Soc.*, 2002, vol. 18, pp. 5222–5230.
- Peter, M.L., Rob, G.H., Hempenius, M.A., and Vancso, J., Electrochemistry of surface-graft stimulus-responsive monolayers of poly(ferrocenyldimethylsilane) on gold, *Langmuir*, 2005, vol. 21, pp. 5115–5123.
- Chuy, C., Ding, J., Swanson, E., Holdcroft, S., Horsfall, J., and Lovell, K.V., Conductivity and electrochemical ORR mass transport properties of solid polymer electrolytes containing poly(styrene sulfonic acid) graft chains, *J. Electrochem. Soc.*, 2003, vol. 150, pp. 271–279.
- Radhi, M.M., Tan, W.T., and Haider, A.J., Electrochemical characterization of the redox couple of Fe(II)/Fe(III) mediated by a graft polymer electrode in nonaqueous electrolyte, *Int. J. Electrochem. Sci.*, 2012, vol. 7, pp. 1–10.
- Xiong, H.-M., Wang, Z.-D., Xie, D.-P., Cheng, L., and Xa, Y.-Y., Stable polymer electrolytes based on polyether-graft ZnO nanoparticles for all-solid-state lithium batteries, *J. Mater. Chem.*, 2006, vol. 16, pp. 1345–1349.
- Idris, M.R. and Quayum, E.M., Preparation of ZnO nanoparticles and its electrochemical studies, *Nano Vision*, 2013, vol. 2, pp. 76–83.
- Ramanavicius, A., Oztekin, Y., Balevicius, Z., Kausaite-Mikstiniene, A., Krikstolaityte, V., Baleviciute, I., Ratautaite, V., and Ramanaviciene, A., Conducting and electrochemically generated polymers in sensor design, *Procedia Eng.*, 2012, vol. 47, pp. 825–828.
- Starowicz, M. and Stypuła, B., Electrochemical synthesis of ZnO nanoparticles, *Eur. J. Inorg. Chem.*, 2008, vol. 6, pp. 869–872.
- Xiong, H.-M., Liu, D.-P., Xia, Y.-Y., and Chen, J.-Sh., Polyether-graft ZnO nanoparticles with tunable and stable photoluminescence at room temperature, *Chem. Mater.*, 2005, vol. 12, pp. 3062–3064.
- Fang, L., Zhang, B., Li, W., Zhang, J., Huang, K., and Zhang, Q., Fabrication of highly dispersed ZnO nanoparticles embedded in graphene nanosheets for high performance supercapacitors, *Electrochim. Acta*, 2014, vol. 148, pp. 164–169.
- Hong, R.Y., Chen, L.L., Li, J.H., Zheng, Y., and Ding, J., Preparation and application of polystyrene-graft ZnO nanoparticles, *Polym. Adv. Technol.*, 2007, vol. 18, pp. 901–909.
- Achilleos, D.S. and Vamvakaki, M., End-graft polymer chains onto inorganic nano-objects, *Materials*, 2010, vol. 3, pp. 1981–2026.
- Li, Sh., Lin, M.M., Toprak, M.S., Kim, D.K., and Muhammed, M., Nanocomposites of polymer and inorganic nanoparticles for optical and magnetic applications, *Nano Rev.*, 2010, vol. 1, pp. 1–19.
- El-Said, W.A. and Choi, J.-W., Electrochemical biosensor consisted of conducting polymer layer on gold nanodots patterned indium tin oxide electrode for rapid and simultaneous determination of purine bases, *Electrochim. Acta*, 2014, vol. 123, pp. 51–57.
- Radhi, M.M., Al-Damlooji, N.K., Radhi Jobayr, M., and Dawood, D.S., Electrochemical sensors of cyclic voltammetry to detect Cd(II) in blood medium, sensors, *Transducers*, 2013, vol. 155, pp. 150–154.
- Tan, W.T., Radhi, M.M., Rahman, M.Z.B., and Kassim, A., Synthesis and characterization of graft polystyrene with acrylonitrile using gamma-irradiation, *J. Appl. Sci.*, 2010, vol. 10, pp. 139–144.
- Tan, W.T., Farhan, Y., and Zulkarnain, Z., Electrochemical reduction of potassium ferricyanide mediated by magnesium diboride modified carbon electrode, *Sensors Transducers J.*, 2009, vol. 104, pp. 119–127.
- Bard, A.J. and Larry, L.R., *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., New York: Wiley, 2000.
- Zhang, J., Electrochemical methods: Fundamental and applications, *Electro Anal. Chem.*, 1972, vol. 331, pp. 945–957.
- Instruction Manual, CV 50W, ver. 2, West Lafayette: Bioanalytical System, 1996.
- Radhi, M.M., Amir, Y.K.A., Alwan, S.H., and Tee, T.W., Electrochemical effect of different modified glassy carbon electrodes on the values of diffusion coefficient for some heavy metal ions, *J. Phys.: Conf. Ser.*, 2013, vol. 431, pp. 1–7.